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Statistics and Discrete Structures

Exercise sheet 3: Sorting

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SS26

Due date: 23.02.2026

Note: Some algorithms may have different names in the literature. Please make sure to refer to the algorithms explained in the lecture notes and in the videos.

Exercise 1 (Sorting algorithms: bubble sort, merge sort, and quick sort | 5 points)
Sort the set $\{6, 69, 9, 53, 7, 2, 65, 37\}$ *using the following algorithms from the lecture notes. Show every step and provide the number of comparisons made.*

(a) *Bubble sort (Algorithm 2.4),*
(b) *Merge sort (Algorithm 2.11),*
(c) *Quick sort (Algorithm 2.15).*

Exercise 2 (Practice with big-O | 5 points)
For each pair of functions below, determine and prove whether it holds $f = \mathcal{O}(g)$, $g = \mathcal{O}(f)$, *both, or none.*

(a) $f(n) = 2n^2 + 3n - 6, \quad g(n) = n^2 - 8n.$
(b) $f(n) = \log_2(n^2), \quad g(n) = \log_3(n)^2.$
(c) $f(n) = n!, \quad g(n) = n^n.$

Exercise 3 (Ω– and Θ–notation | 5 points)
Similar to the lecture we define

- $f(x) = \Omega(g(x))$ *if there exist constants* $c, C > 0$ *such that*
$$\forall x > C : f(x) \geqslant c \cdot g(x),$$
- $f(x) = \Theta(g(x))$ *if* $f(x) = \mathcal{O}(g(x))$ *and* $f(x) = \Omega(g(x))$.

Prove that

(a) $f(x) = \mathcal{O}(g(x)) \iff g(x) = \Omega(f(x)),$
(b) $f(x) = \Theta(g(x)) \iff g(x) = \Theta(f(x)),$
(c) $f(x) = \mathcal{O}(g(x)) \wedge g(x) = \mathcal{O}(f(x)) \iff f(x) = \Theta(g(x)).$

Exercise 4 (Stability | 5 points)
For this exercise, unlike so far, let us now consider inputs which may contain repetitions. Then, a sorting algorithm is said to be ***stable*** *if the order of repeated elements is preserved after sorting.*

Let us define it formally. Let $z_1, z_2, \dots, z_n \in \mathbb{R}$ *be* n *real numbers to sort. Considering possible repetitions, a sorting algorithm returns a permutation of* $\{1, 2, \dots, n\}$*, namely* $\pi_1, \pi_2, \dots, \pi_n$*, such that* $z_{\pi_1} \leqslant z_{\pi_2} \leqslant \cdots \leqslant z_{\pi_n}$*. We call a sorting algorithm* ***stable*** *if*

$$\forall i, j \in \{1, 2, \dots, n\} : z_i = z_j \ \wedge \ i < j \implies \pi^{-1}(i) < \pi^{-1}(j).$$

Analyze the stability of bubble sort, merge sort, and quick sort, exactly as described in the lecture notes, with the following assumptions:

- *In bubble sort, the maximum* z_{π_k} *is selected as the left-most element which is larger or equal than the rest of the set* S_k*.*
- *In quick sort, the pivot* x *is chosen as the first element in* S*, and the elements equal to it are grouped together with the elements larger than it (i.e.,* $\{z_{\pi_{m+1}}, z_{\pi_{m+2}}, \dots, z_{\pi_n}\}$*).*